

Joseph W. Choi

Ph.D. Student

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Education

Massachusetts Institute of Technology , Cambridge, MA	2030
Ph.D. in Chemical Engineering	
Thesis Advisor: Sungho Shin	
University of Connecticut , Storrs, CT	2025
B.S.E. in Chemical Engineering	

Research Interests

nonlinear programming; dynamic optimization; GPU computing

Appointments

Research Assistant	2025–2030
Department of Chemical Engineering, Massachusetts Institute of Technology, Cambridge, MA	
Supervisor: Sungho Shin	
Undergraduate Research Assistant	2023–2025
Department of Chemical and Biomolecular Engineering, University of Connecticut, Storrs, CT	
Supervisor: Matthew D. Stuber	

Honors and Awards

2nd Place, Undergraduate Student Poster Competition (Computing and Process Control III), AIChE	2024
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Mentoring Experience

Massachusetts Institute of Technology , Cambridge, MA	
Rachel Wu (UROP)	Fall 2026

Teaching Experience

University of Connecticut , Storrs, CT	
CHEG 3124 Heat and Mass Transfer , Undergraduate Teaching Assistant	Spring 2025
CHEG 3145 Chemical Engineering Numerical Analysis , Undergraduate Teaching Assistant	Fall 2024

Publications

Journal Publications

- [J1] D. Alston*, **J. W. Choi***, P. Xu, and M. D. Stuber. EOptInterface.jl: Equation-oriented optimization interface in Julia. *Industrial & Engineering Chemistry Research*, 65(26):13847–13861, 2026. doi:[10.1021/acs.iecr.6c00186](https://doi.org/10.1021/acs.iecr.6c00186).
*Shared co-first authorship.

Conference Publications

- [C1] **J. W. Choi** and S. Shin. Revisiting simultaneous methods for dynamic optimization in the GPU era. In *Foundations of Computer Aided Process Operations / Chemical Process Control (FOCAPO/CPC)*, 2027, [2607.12201](#). Accepted.

Presentations

- [T1] **J. W. Choi** and M. D. Stuber. Automatic generation of reduced-space models for faster global optimization in Julia. Poster, AIChE Annual Student Conference, San Diego, CA, Oct. 2024. 2nd Place, Computing and Process Control III.

Software

- [S1] **ExaModelsCollocation.jl** (Main developer)
- tool for transcribing differential algebraic equations via orthogonal collocation in ExaModels.jl
 - <https://github.com/madsuite-org/ExaModelsCollocation.jl>
- [S2] **ExaModelsPEtab.jl** (Main developer)
- tool for solving PEtAb parameter estimation models with ExaModels.jl
 - <https://github.com/mit-shin-group/ExaModelsPEtab.jl>
- [S3] **ParametricNLPModels.jl** (Developer)
- data structure for computing parametric sensitivities in nonlinear programming models
 - <https://github.com/madsuite-org/ParametricNLPModels.jl>
- [S4] **ExaModels.jl** (Contributor)
- a sparse automatic differentiation and algebraic modeling tool
 - <https://github.com/madsuite-org/ExaModels.jl>
- [S5] **MadNLP.jl** (Contributor)
- a nonlinear programming solver
 - <https://github.com/madsuite-org/MadNLP.jl>
- [S6] **EOptInterface.jl** (Main developer, 2024–2025)
- abstraction layer for optimizing componentized equation-oriented models
 - <https://github.com/PSORLab/EOptInterface.jl>

References

Provided upon request.

Last updated: September 26, 2026